

Ruihan WU

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EDUCATION

Southern University of Science and Technology (SUSTech)

Shenzhen, China

Major in Information Engineering

Sept 2023—Jun 2027 (Expected)

- **GPA:** 3.93/4.0, **Rank:** 1/48

University of Pennsylvania

Philadelphia, The United States

International Guest Student Program

Jan 2026—May 2026

- **GPA:** 4.0/4.0

Johns Hopkins University

Baltimore, The United States

Summer Research Internship Program

June 2026—September 2026(Expected)

- **Major Courses:** Robot Perception and Intelligence (98), Information theory (99), Communication Principles (96), Linear Algebra (95), Signals and Systems (94), Deep Learning (99), Digital Circuits (98), Engineering Electromagnetic Field Theory (98), Probability Theory and Mathematical Statistics (95)

ACADEMIC PROJECTS

Provably Optimal MARL for Safety-Filtered Quadruped Zero-Sum Games

Baltimore, The United States

Alliance AI Lab, Johns Hopkins University

2026.6—present

- Implemented a boundary-aware Control Barrier Function (QCBF) safety filter for quadrupedal locomotion, enforcing field-boundary constraints via a value-based least-restrictive filtering layer.
- Built a GPU-accelerated massively-parallel training environment implementing the ISAACS adversarial safety-game protocol on the mjlabs/MuJoCo Warp simulation stack for scalable multi-agent policy training.
- Designed and implemented a two-player zero-sum "touchdown" game trained via adversarial reinforcement learning, extending the PORL framework to a filtered Markov game setting.

Physical Residual Distillation for AUV Reinforcement Learning Control

Philadelphia, The United States

University of Pennsylvania

2025.12—2026.03

- Proposed a high-fidelity hybrid simulation platform for Autonomous Underwater Vehicles (AUVs) Reinforcement Learning training by coupling rigid-body physics (MuJoCo) with a data-driven hydrodynamic model.
- Implemented a residual learning pipeline by collecting high-fidelity hydrodynamic data from Stonefish and training a MLP to predict complex non-linear fluid interactions (e.g., added mass, damping) with low latency.
- Published "Deep Hydro-Sim: Structured Residual Hydrodynamics for the RL-Based AUV Control" as the sole first author at the 2026 IEEE ICMA and was invited to deliver an oral presentation at the conference.

Design of Underwater Navigation and Control System for Coaxial Twin-Propeller ROV Based on ROS2 and Visual SLAM

Shenzhen, China

Guo Lab, SUSTech, supervised by Prof. Shuxiang Guo

2025.10—2026.1

- Integrated a stereo camera and other auxiliary sensors to provide high-precision position data in underwater environments, supporting map construction for the vision-inertial based SLAM system.
- Developed the SLAM system based on the ROS2 framework between different devices; deployed the ORB_SLAM3 algorithm for underwater environment mapping.

Design of Motion Control System based on TurtleBot4

Shenzhen, China

Robotics Intelligence & Perception Course Project

2025.9—2025.12

- Integrated robotic arms with the TurtleBot4 platform to realize collaborative planning of mobile locomotion and grasping movements, achieving seamless coordination between base movement and end-effector operation.
- Leveraged an RGB camera to implement target recognition and pose estimation via OpenCV.

Design of Automatic Surface Water Cleaning Unmanned Boat

Shenzhen, China

"Shokz Cup" Global Outstanding Innovative Talent Summer School

2025.6.23-2025.7.13

- Designed a vision-based motion control target tracking system, integrating components such as YOLOv8n object detection, PX4, and servo motors.

- Designed and implemented robotic arm gripping mechanisms on both sides of the unmanned vessel, which work in conjunction with the decision-making system to perform waste collection functions.

Laser Automatic Aiming Device

Shenzhen, China

National Undergraduate Electronic Design Competition

2025.7.30-2025.8.3

- Designed a laser-guided automatic target-following vehicle based on STM32 and K230. Built from scratch in 4 days to enable the robot to execute a complete task as the team leader.
- Awarded the National Second Prize, achieving the best historical performance in SUSTech history.

Research on Smoke Grenade Deployment Strategies for Drone Swarms

Shenzhen, China

National Undergraduate Mathematical Modeling Contest

2025.9.02-2025.9.06

- Established a dynamic smoke diffusion and laser transmittance model based on the Beer-Lambert law, quantitatively evaluating the interference efficiency by calculating the extinction coefficient and mass concentration distribution of the aerosol cloud.
- Implemented a Particle Swarm Optimization (PSO) algorithm to solve for the optimal tactical parameters, successfully maximizing the effective screening area and duration against laser guidance systems.

HONORS & AWARDS

Second Prize, SUSTech Outstanding Student Scholarship, 2023-2024 & 2024-2025

Outstanding Student, SUSTech, 2024&2025

Silver Prize, “Shokz Cup” Global Outstanding Innovative Talent Summer School

National Second Prize, National Undergraduate Electronics Design Contest

Provincial First Prize & National Second Prize, National Undergraduate Mathematical Modeling Contest

Silver Prize, “Foreign Language Press · China Talent Cup” English Comprehensive Proficiency Competition

Third Prize, “Foreign Language Press · China Talent Cup” English Listening Competition

SKILLS

Language: English (Fluent, TOEFL 109), Chinese (Native) | **Technical:** C, Python, bash, ROS, Reinforcement Learning, Convex Optimization, Deep Learning